

CV



Ali Değer Özbakır

Assistant
Professor,
Computer
Science/AI,
Open
Universiteit.
Background in
geophysics,
machine
learning, and
software
engineering.

- ✉ Email
- 🔍 Google Scholar
- 🆔 ORCID
- 🐙 GitHub
- 🌐 LinkedIn

EDUCATION

- PhD in Geophysics, Utrecht University, Netherlands (2019) — dissertation: "Evolving plate boundaries in the Aegean-Anatolian region"
- MSc in Solid Earth Geophysics, Eurasia Institute of Earth Sciences, Istanbul Technical University
- BSc in Geological Engineering, Istanbul Technical University
- BSc in Mining Engineering, Istanbul Technical University (ranked first in class)

WORK EXPERIENCE

2025–PRESENT

Assistant Professor, Computer Science/AI

🏛️ *Open Universiteit, Heerlen, Netherlands*

- Teach Time Series Analysis and Forecasting; co-developed the MSc course Software Engineering and AI
- Designed the system architecture for MAS4TE, a multi-agent reasoning platform for energy trading, including smart contracts for battery storage trading
- Tech lead for GRINS, an end-to-end pipeline (with the University of Verona) extracting and classifying green-innovation claims from corporate annual reports
- Supervise PhD candidate Yu Sheng on AI-driven energy optimization (MAI-HOME project, expected completion 2027)

2023–2025

Postdoctoral Researcher

 *Open Universiteit and University of Twente, Netherlands*

- Time-series forecasting (RNN/LSTM/CNN/Transformers) for occupancy inference and energy optimization from sparse IoT data
- Executed the full technical implementation of GO-KIT, a cross-border NL–BE smart-home energy pilot, including running LSTM networks in the test house
- Contributed software development to a private Open Universiteit/APG collaboration on AI-based detection of corporate greenwashing
- Extended time-series and index-based heuristics to optical Earth observation imagery
- Built the API and core functionalities (model inference, iterative mosaicking with uncertainty) for the NWO-funded “Landslide Hunter” platform for automated landslide detection and mapping; co-authored 3 conference papers on it

2021–2023

Data Scientist & Software Developer, Temblor Inc.

 *San Francisco, USA*

- Led a global spatiotemporal earthquake activity-rate model for insurance/reinsurance risk assessment
- Built a parallel variable-bandwidth KDE framework, achieving roughly 20x resolution improvement
- Integrated hazard/fragility models into production decision-support APIs

2020–2021

Postdoctoral Researcher

 *Kandilli Observatory and Earthquake Research Institute, Istanbul, Turkey*

- Secured competitive funding for a Bayesian probabilistic thermal model (tomography, spectral analysis) and taught a graduate course on Plate Tectonics and Crustal Deformation

2014–2018

Program Manager, Education & Digital Initiatives, TEMA Foundation

 *Istanbul, Turkey*

- Built a national e-learning platform for environmental education and a citizen-science biodiversity app
- Managed a ~\$500K annual budget and a team of 5; secured €177K in EU competitive funding

2012–2014

Research Engineer, Kandilli Observatory

 *Istanbul, Turkey*

- Optimized GNSS network coverage via clustering (k-means, hierarchical), improving processed stations by over 20%
- Non-linear regression analysis on GNSS time series

SKILLS

Programming

Python

C/C++

MATLAB

FORTRAN

Solidity

bash

Machine learning

TensorFlow/Keras

scikit-learn

Transformers/LLMs

Prophet

spaCy

Chroma

Data & web

PostgreSQL/SQL

SQLAlchemy

Flask

FastAPI

HTML/CSS/JS

Azure Document Intelligence

GIS/geospatial

QGIS

GeoPandas

rasterio

Languages

English (fluent)

Dutch (A2)

Turkish (native)

PUBLICATIONS AND TEACHING

The full publication list (including earlier geophysics work and work under review/in preparation) is on the [Publications](#) page or [Google Scholar](#). Teaching and supervision are on the [Teaching](#) page.

SERVICE AND LEADERSHIP

- Convener, vEGU21 / EGU22 — “Active tectonics and geodynamics of the Eastern Mediterranean”
- Reviewer for *Earth and Planetary Science Letters*, *Nature Communications Earth & Environment*, *Mediterranean Geoscience Reviews*